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DATA NETWORKS

Lab 2: Linux

Lab Objectives:

- Obtain basic network information
- Network commands used to retrieve network-specific information in Linux
- Compare Linux and Windows basic network information and commands

Instructions:

- Use the workstations in the FCI labs or your laptop to complete the lab work.
- Ensure that your workstation is connected to the NUST Network.
- You may also use your laptop connected to the NUST Network.
- If you are using your computer, **download and install Oracle VirtualBox** (it is advisable to use the same version that is installed on the computers in the FCI labs)
- **Download the Linux OS image (Ubuntu)** from the following sites:
  - \\172.21.253.166\isnotes\STUDENTS SOFTWARE 2023
  - https://releases.ubuntu.com/jammy/ubuntu-22.04.5-desktop-amd64.iso
- Set your **surname** as your **username** and your **password** as your **first name**.
- Import it into the Oracle VirtualBox and install the OS.
- Start up the Ubuntu virtual machine.
- Make sure your Linux virtual machine has network connectivity by configuring it.

Virtual machine specifications:

Hard Disk Size: 50 GB (Minimum, allocate more if required)

Memory Size (RAM): 2048 MB (minimum, allocate more if you can)

Processors: 2 (allocate more if required)

Network: NAT (network connectivity)

(If you are using a personal laptop, configure settings according to your laptop specification)

Tools / Equipment

- Oracle VirtualBox (https://www.virtualbox.org/wiki/Downloads)
- Ubuntu ISO

Section 1 [25 Marks]

1) Install the Linux Operating System and ensure that the VM has internet connectivity. *(See instructions and specifications above).*

2) Open the terminal (command line) and record the following information:

   ***Hint**: check your terminal*

   i. Identify your login name: _____ (1)

   ii. Identify your hostname: _____(1)

   iii. Check the available network cards. Determine if they are active or inactive and if they are physical or logical devices. Use the **ifconfig -a** command _____(3)

Hint: There can be only one network card that is both active and physical. Other active network cards are logical devices (e.g., a loopback, or a VMWare tunnel), or are disabled (e.g., if you are on wireless Ethernet but also have a wired Ethernet card).

No	Network Adapter	Physical/Logical	Active/Inactive

2. What is the difference between a physical and a logical adaptor ? (2)

_____

3. Record the following network information/ settings.

a) Record the IP address and subnet mask. (2)

_____

b) Record the Default Gateway. Use the command: ***route -n .*** Use the manual page for route (command: **man route**) (2)

_____

Man is a command that provides the description of the command and how it can be used.

c) Record the DNS Server IP Address. Use the command: ***cat /etc/resolv.conf*** *(2)*

_____

d) Is DHCP enabled? Record the DHCP Server IP Address, use the commands:

sudo dhclient -v = requests a dhcp lease from the DHCP server

sudo cat /var/lib/dhcp/dhclient.leases **=** displays the dhcp lease file *(2)*

_____

e) Determine if all DNS servers in ***/etc/resolv.conf*** have been assigned through the DHCP server (explain) (3)

_____

_____

_____

f) Test your machine's connectivity to the NUST website, troubleshoot if there is any connectivity errors. (2)

_____

g) Which command will show the manual page for ping command? (2)

_____

h) How can you use ping to determine if the DNS server is working? (3)

_____

Section 2 [13 Marks]

Part A

1. Create a new Loopback interface: (2)
   a) Configure the name as **Loopback 10**. Use the commands:

      sudo ip link add name loopback10 type dummy

      _____

   b) Configure the IP address **127.0.0.10** with subnet mask **255.255.255.0** to this interface. *Enable the loopback interface* (2). Use the commands:

      sudo ip addr add 127.0.0.10/24 dev loopback10
      sudo ip link set loopback10 up

      _____
      _____

   c) Create route 127.0.0.11 on the loopback interface.
      Use the command **sudo ip route add 127.0.0.11/32 dev lo:10**

      _____
      _____

   d) Verify that a route has been created for this host using the command **route -n.** Also verify that the loopback interface has been created using **ifconfig -a** (2)

      _____

   e) What is a loopback interface, what is its' function and why would there be a need to create a loopback interface? (3)

      _____
      _____
      _____
      _____

   f) Why would you create/add a route to your systems routing table? What is a routing table ? (2)

      _____
      _____
      _____

   g) Run the command to translate the IP 8.8.8.8 to its' hostname? (1)

      _____
      _____
      _____

Part B: Self-study [12]

1. Find the network information/ settings you recorded in Section 1, Question 2 (**a–f)** without using the terminal. Write down the steps taken. (2)

_____

_____

_____

_____

_____

2. Open the terminal, type the command: ***arp -a*** and list the internet addresses with their associated physical addresses. (2)

_____

_____

_____

_____

_____

_____

3. ***Ping*** the following IP addresses:

   i. Fellow student's IP address
   ii. Your Windows machine IP address
   iii. IP Address: 8.8.8.8

After each ping (above), type the arp -a command again and record newly learned MAC addresses, if any.

List the new internet addresses with their associated physical addresses that your computer just learned. (3)

i. _____

ii. _____

iii. _____

a) Did your computer learn all the IP addresses pinged? If NO, list the IP addresses that were pinged but not learned and explain why your computer learned/ did not learn some of these IPs (3)

b) Identify the the MAC address of the gateway? (2)

End of Lab